

PASSAU, Germany

WMO: 108930

Lat: 48.583N

Lon: 13.467E

Elev: 409

StdP: 96.51

Time Zone: 1.00 (EUC)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.7	(c) -11.5	(d) -16.9	(e) 0.9	(f) -12.6	(g) -14.2	(h) 1.2	(i) -9.5	(j) 9.9	(k) 3.6	(l) 8.2	(m) 2.2	(n) 0.7	(o) 40	(p) 0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.9	(c) 29.1	(d) 18.8	(e) 27.4	(f) 18.3	(g) 25.8	(h) 17.6	(i) 20.1	(j) 27.1	(k) 19.2	(l) 25.4	(m) 18.4	(n) 24.0	(o) 1.8	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	13.3	22.1	17.0	12.7	21.2	16.3	12.2	20.5	59.2	27.0	56.2	25.6	53.6	24.1	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.9	(b) 5.5	(c) 4.6	(e) -16.3	(f) 32.5	(g) 4.8	(h) 2.0	(i) -19.8	(j) 34.0	(k) -22.6	(l) 35.2	(m) -25.3	(n) 36.3	(o) -28.7	(p) 37.8		
(a) 6.9	(b) 5.5	(c) 4.6	(e) -16.5	(f) 21.9	(g) 4.7	(h) 0.8	(i) -19.9	(j) 22.5	(k) -22.7	(l) 23.0	(m) -25.3	(n) 23.4	(o) -28.8	(p) 24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-1.8	-1.0	4.0	8.1	13.3	15.6	18.5	18.1	14.3	9.3	2.9	-0.1
	DBStd	8.11	5.02	4.64	4.08	3.55	3.76	3.27	3.08	3.29	2.97	3.64	3.37	3.82
	HDD10.0	1543	366	308	189	78	15	1	0	0	2	57	213	314
	HDD18.3	3694	624	541	444	307	159	94	37	45	127	280	462	573
	CDD10.0	988	0	0	3	21	116	168	263	252	129	36	1	0
	CDD18.3	97	0	0	0	0	2	12	42	38	4	0	0	0
	CDH23.3	1013	0	0	0	1	37	133	428	370	44	1	0	0
	CDH26.7	234	0	0	0	0	1	18	113	99	3	0	0	0
Wind	WSAvg	1.6	1.6	1.7	1.9	2.0	1.9	1.6	1.5	1.4	1.4	1.5	1.4	1.5
Precipitation	PrecAvg	951	75	66	73	51	89	100	110	98	75	67	71	77
	PrecMax	1320	173	146	202	103	152	153	189	264	146	169	146	168
	PrecMin	664	9	13	14	3	14	35	39	23	29	17	0	20
	PrecStd	137	43	36	51	25	40	35	41	50	34	38	37	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	13.2	19.2	22.5	26.0	28.6	32.0	32.4	26.5	22.7	12.9	10.1
		MCWB	6.5	7.7	11.2	12.8	15.7	18.1	20.1	20.0	18.4	15.8	8.6	8.1
	2%	DB	7.0	9.6	15.4	20.1	24.4	26.8	29.6	29.1	24.6	19.9	11.0	7.8
		MCWB	5.4	6.2	9.2	12.0	15.1	17.7	19.2	19.1	17.4	14.1	8.6	6.3
	5%	DB	5.5	6.7	12.8	17.7	22.7	24.8	27.6	27.3	22.7	17.2	9.1	6.5
		MCWB	4.4	4.4	7.8	10.5	14.2	16.9	18.4	18.6	16.6	12.9	6.9	5.4
	10%	DB	4.1	4.6	10.2	15.5	20.9	22.6	25.8	25.3	20.4	15.0	7.5	5.0
		MCWB	3.1	3.0	6.5	9.5	13.5	15.7	17.8	17.9	15.1	11.9	5.9	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	8.5	11.8	14.1	17.7	19.9	21.5	21.3	19.4	16.5	10.5	8.1
		MCDB	8.5	11.5	18.0	20.3	23.3	26.1	29.2	29.8	24.7	20.6	11.9	9.8
	2%	WB	5.7	6.6	9.7	12.4	16.1	18.7	20.1	20.1	18.3	14.8	9.0	6.6
		MCDB	6.8	9.3	13.6	18.9	21.8	24.6	27.2	27.5	23.2	18.7	10.6	7.6
	5%	WB	4.5	4.7	8.4	11.2	15.0	17.6	19.2	19.2	17.1	13.4	7.3	5.6
		MCDB	5.3	6.6	11.9	16.2	20.7	23.0	25.6	25.6	21.2	16.3	8.6	6.5
	10%	WB	3.2	3.2	7.1	10.0	14.1	16.5	18.4	18.3	15.9	12.1	6.2	4.0
		MCDB	4.1	4.5	9.4	14.4	19.4	21.4	24.1	23.7	19.2	14.6	7.2	4.9
Mean Daily Temperature Range	5% DB	MDBR	5.1	7.0	8.4	9.8	10.5	9.7	10.9	10.6	9.8	8.7	5.6	4.5
		MCDBR	6.5	10.4	13.6	14.7	14.0	13.7	14.8	14.8	13.8	12.8	8.2	6.5
		MCWBR	5.1	6.7	7.4	7.5	6.4	6.3	6.0	6.2	7.0	7.4	5.8	5.2
	5% WB	MCDBR	6.0	9.5	12.0	12.9	12.2	12.2	13.0	13.0	12.3	11.5	7.2	6.2
		MCWBR	5.0	6.5	7.0	7.0	5.8	6.1	5.8	5.8	6.5	7.0	5.4	5.1
Clear-Sky Solar Irradiance	taub	0.298	0.323	0.363	0.397	0.403	0.406	0.415	0.408	0.381	0.351	0.324	0.297	
	taud	2.391	2.345	2.301	2.251	2.269	2.295	2.271	2.307	2.352	2.420	2.450	2.438	
	Ebn at Noon	767	826	846	851	858	857	842	831	817	784	730	723	
	Edh at Noon	67	88	108	126	129	126	128	118	102	81	64	57	
All-Sky Solar Radiation	RadAvg	1.00	1.92	2.98	4.32	5.03	5.55	5.35	4.69	3.35	2.09	1.06	0.75	
	RadStd	0.17	0.26	0.37	0.55	0.60	0.48	0.50	0.48	0.44	0.23	0.12	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (87 neighbors)		+0.46	N/A	N/A	N/A	+0.37	+0.39	N/A	N/A	+62	+26

Nomenclature: See separate page